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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] MAY-JUNE 2026

Paper Code: BS-106

Subject: Applied Physics-II

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated. Assume missing data if any.

Q. No.	Question	Marks
Q1	Attempt any four of the following questions:	(4×5=20)
(a)	Using Heisenberg's uncertainty principle, prove that electrons do not exist in the nucleus.	(5)
(b)	For the relativistic case, show that the group velocity of a wave packet is always equal to the velocity of the moving material particle.	(5)
(c)	Compute the number of ways according to MB, BE, and FD statistics in which two particles are to be distributed in three energy states.	(5)
(d)	Discuss the Fermi distribution function and its behavior at different temperatures.	(5)
(e)	What is a Hexagonal Close-Packed (HCP) structure? Deduce the c/a ratio for an HCP lattice.	(5)
(f)	Sodium Chloride (NaCl) crystallizes in an FCC structure. Calculate the lattice constant. (Given: Molecular Weight of NaCl = 58.45 kg/kmol, density = 2180 kg/m ³).	(5)
(g)	Define the effective mass of an electron. Obtain an expression for the effective mass of an electron moving in a periodic potential.	(5)
(h)	Differentiate between P-type and N-type semiconductors with suitable examples.	(5)
UNIT - I		
Q2 (a)	State de-Broglie hypothesis. How did Davisson and Germer demonstrate de-Broglie waves in the laboratory through an experiment?	(5)
(b)	If the eigen wave function of a particle's state is $\psi = A \sin 2x$, normalize the wave function in the limit $-\pi/2$ to $+\pi/2$. Also, calculate the probability of finding the particle between 0 and $\pi/4$.	(5)
OR		
Q3 (a)	Write Schrödinger equation for a particle trapped in a one-dimensional infinite potential well. Set up eigen values and eigen functions. Show your results graphically.	(7)
(b)	Discuss the zero-point energy of a quantum harmonic oscillator and show that the energy levels are equally spaced.	(3)

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Q. No.	Question	Marks
UNIT - II		
Q4 (a)	List the salient features of black body spectra. Describe how Classical statistics failed to explain the Black Body spectrum and Bose-Einstein statistics correctly explained it.	(7)
(b)	Verify that the rms speed of an ideal gas molecule is 9% greater than its average speed.	(3)
OR		
Q5 (a)	Define Fermi energy and prove that the average energy of fermions for a completely degenerate electron gas is given by (3/5) E_F .	(5)
(b)	Assuming the Sun's temperature to be 5902 K, calculate the wavelength corresponding to maximum emission of the Sun. (Given: Wien's displacement constant $C = 2.892 \times 10^{-3} \text{ m}\cdot\text{K}$).	(3)
(c)	Write a short note on a dying star.	(2)
UNIT - III		
Q6 (a)	Define Miller Indices. Derive an expression for spacing between two adjacent Miller planes of a simple cubic lattice. Also, draw (011), (121), and (100) planes in a cubic lattice.	(5)
(b)	Derive Bragg's law of X-ray diffraction and discuss Laue's method for crystal structure analysis.	(5)
OR		
Q7 (a)	What do you understand by point defects in solids? Derive an expression to show that the number of Frenkel defects at a given temperature is proportional to ($N \cdot N_i$)^{1/2} , where N and N_i are the number of atoms and interstitial sites respectively.	(6)
(b)	Calculate the glancing angle on a cube of rock salt ($a = 0.2814 \text{ nm}$) corresponding to 2 nd order diffraction maxima for X-rays of wavelength 0.0710 nm.	(4)
UNIT - IV		
Q8 (a)	Outline the formation of allowed and forbidden bands using the Kronig-Penney Model of an electron in a periodic potential and hence draw the E vs k curve.	(6)
(b)	What are Brillouin Zones? Describe Brillouin Zones for 1-D and 2-D lattices with suitable diagrams.	(4)
OR		
Q9 (a)	Describe forward and reverse biasing in a tunnel diode with the help of energy band diagrams, draw its V-I characteristics and explain the negative resistance region and its applications.	(7)
(b)	Explain the working of a Light Emitting Diode (LED).	(3)

**** End of Question Paper ****